WORKSHOP ACTIVITIES

Curriculum Modules

The camp curriculum is organized into a sequence of six modules designed to progressively introduce participants to core computing concepts. The modules follow a scaffolded structure, beginning with visual programming and interactive game design, transitioning to physical computing with Raspberry Pi, and culminating in hands-on Internet of Things (IoT) applications. This progression enables learners to gradually connect abstract programming concepts with tangible, real-world systems.

Module 1: Introduction to Programming (Scratch) (Day 1)

Activities and Games:
- Pop the Balloon
- Tell Your Story

Skills Covered:
- Inputs and outputs, variables, and arithmetic operations
- Control structures (sequence, loops, conditionals)
- Event handling (keyboard and mouse input)

Challenges:
- Design a "Pop the Balloon" game for family members
- Create a storytelling game introducing oneself or one's community

---

Module 2: Interactive Game Design (Day 2)

Activities and Games:
- Let's Fly (Version 1)
- Let's Fly (Version 2)

Skills Covered:
- Control structures (repetition, nested loops, conditionals)
- Keyboard event handling (left, right, up, and down arrows)
- User interaction, animation, sprite collision detection, and multitasking

---

Module 3: Introduction to Raspberry Pi (Day 3)

Activities:
- Astro Pi in Space

Skills Covered:
- Raspberry Pi fundamentals
- Hardware awareness and computing environment
- GPIO input and output concepts

Module 4: Guest Speakers – IT Professionals (Day 3)

Activities:
- Virtual field trip to Choctaw Nation IT Headquarters
- Guest presentations: "A Day in the Life of IT Professionals"

Discussion and Q&A Topics:
- Work environments and professional roles
- Required skill sets and challenges
- Career pathways and opportunities

Module 5: Physical Computing (Day 4)

Activities and Games:

• Display and scroll "Hello World" on the 8×8 LED matrix
– *Challenge:* Create an interactive "Hello John" application
• Control a sprite using the Sense HAT joystick
– *Challenge:* Create a "Catch the Apple" or "Snake" game
• Display temperature data on the LED matrix
– *Discussion:* Understanding the role of mathematics in sensor data
– *Challenge:* Create a weather station application
• Accelerometer-based motion detection

Skills Covered:

• Scratch Sense HAT extension blocks
• Sense HAT hardware components (LED matrix, temperature, humidity, barometric pressure, gyroscope, accelerometer, magnetometer, and joystick)
• Programming hardware input/output interactions using Scratch

---

Module 6: IoT Integration (Peak Learning) (Day 4)

Activities and Games:

• Game 3: "Don't Shake Me!"

Skills Covered:

• Text-to-speech integration
• Real-world sensor data interaction
• Core concepts of Internet of Things (IoT) systems
• Use of the Cayenne IoT platform

IoT concepts were introduced by connecting sensor inputs (e.g., temperature, motion) from the Raspberry Pi and Sense HAT to interactive with Scratch programs, allowing students to observe real-time physical data influencing digital behavior.

Daily Instructional Flow

As shown in Table 2 (Appendix), the daily instructional flow was designed to balance short lectures, hands-on activities, and scheduled breaks to sustain student engagement and attention. The camp followed a structured four-day progression that integrated introductory programming, interactive game design, and physical computing. Each morning began with participant check-in, followed by Scratch-based game activities on Monday and Tuesday to introduce core programming concepts such as events, animation, and control logic. On Wednesday, instruction shifted toward hardware-focused learning through an introduction to Raspberry Pi, complemented by virtual field trips and guest speakers from IT professions. Thursday emphasized physical computing and Internet of Things (IoT) concepts, combining Scratch with Raspberry Pi and Sense HAT through interactive projects such as joystick control, sensor-based displays, and the "Don't Shake Me" activity. Short breaks were incorporated throughout each day, and volunteers assisted with check-in and check-out to streamline logistics and support instructional flow.